

FEMU: Reproducing ZNS Performance Characteristics

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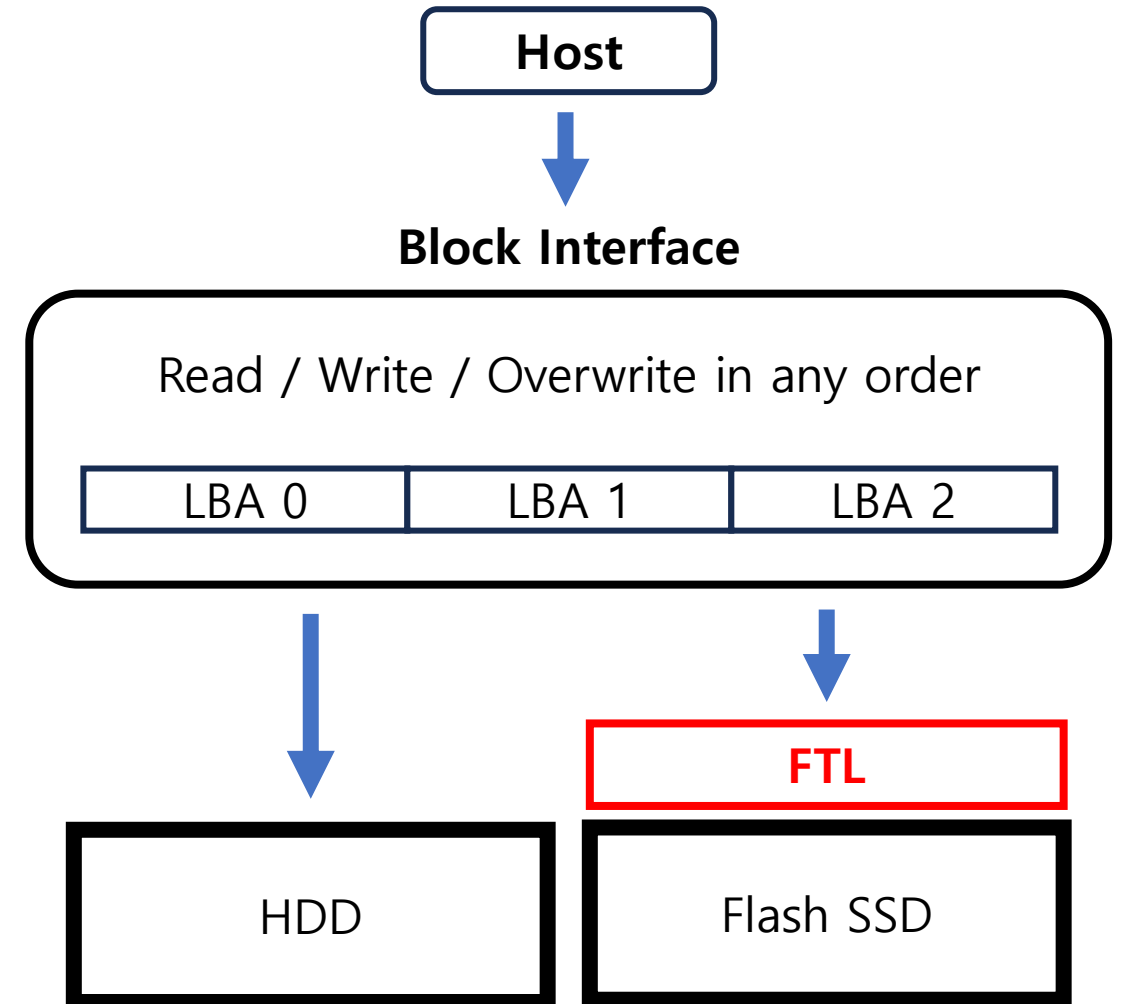
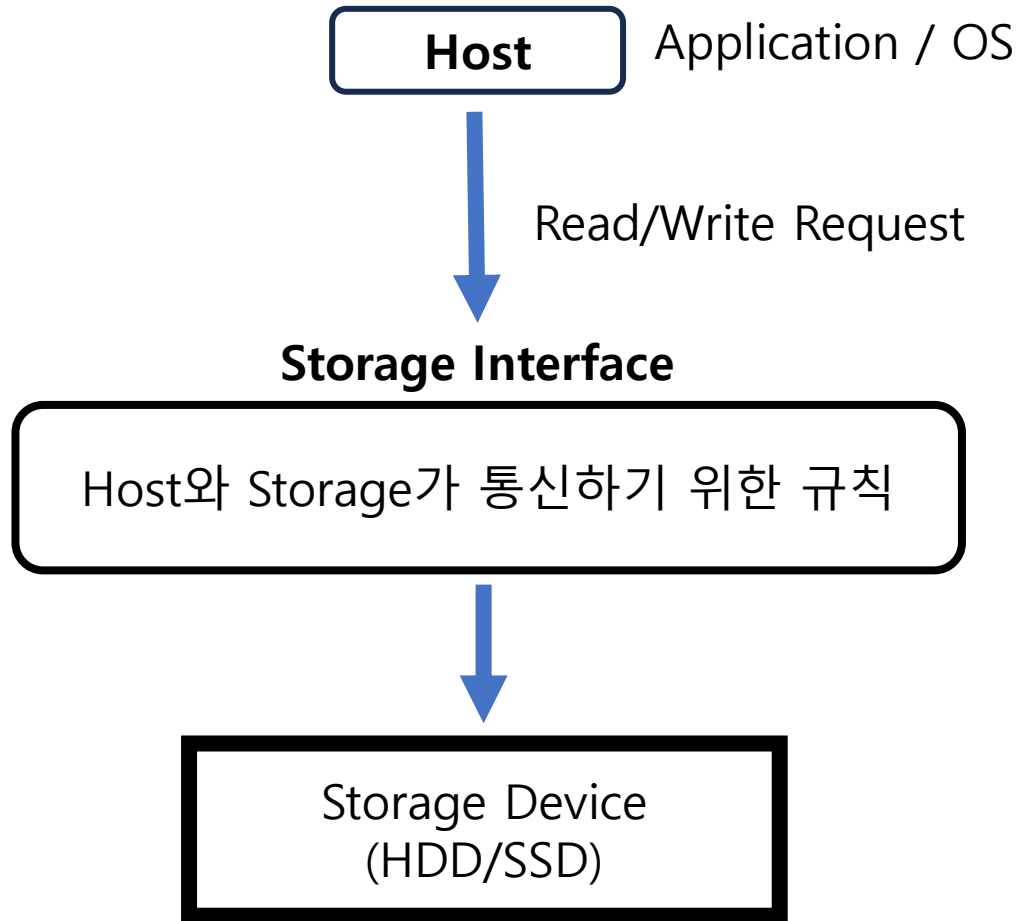
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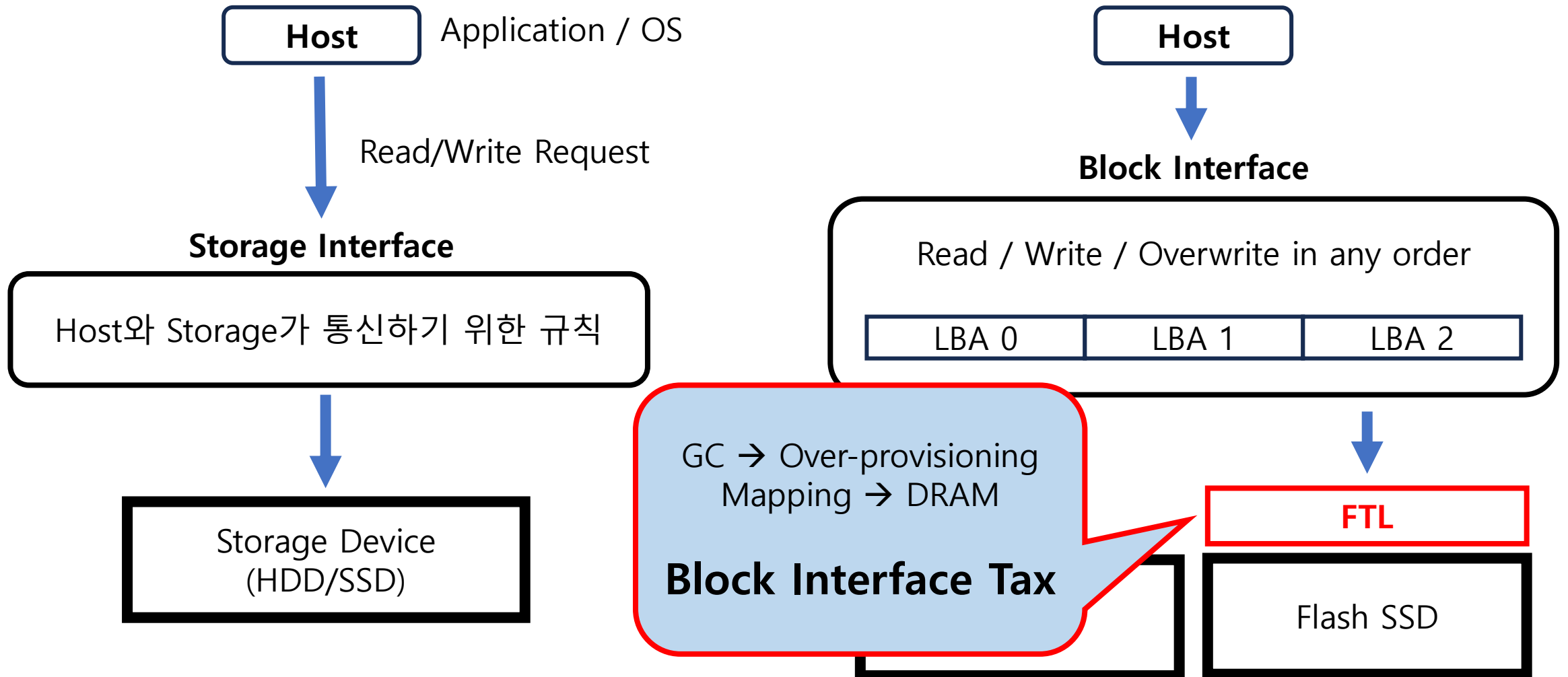
Motivation

- 기존 논문 → **ZNS: Avoiding the Block Interface Tax for Flash-based SSDs**
 - 동일한 SSD Hardware에서 Host의 Write Target을 증가시키며 실제 Write Throughput을 측정
 - Block SSD는 Write Throughput이 특정 지점에서 제한
 - ZNS SSD는 높은 Write Target까지 처리 가능
- 실험 목표
 - 오픈소스 SSD Emulator인 FEMU에서 BBSSD와 ZNSSD를 구성하여 논문의 Figure 5(a) 실험을 재현
 - 제한된 환경에서도 **Block Interface, ZNS** 간 성능 차이와 GC 제거 효과가 동일하게 나타나는지 검증하고자 함

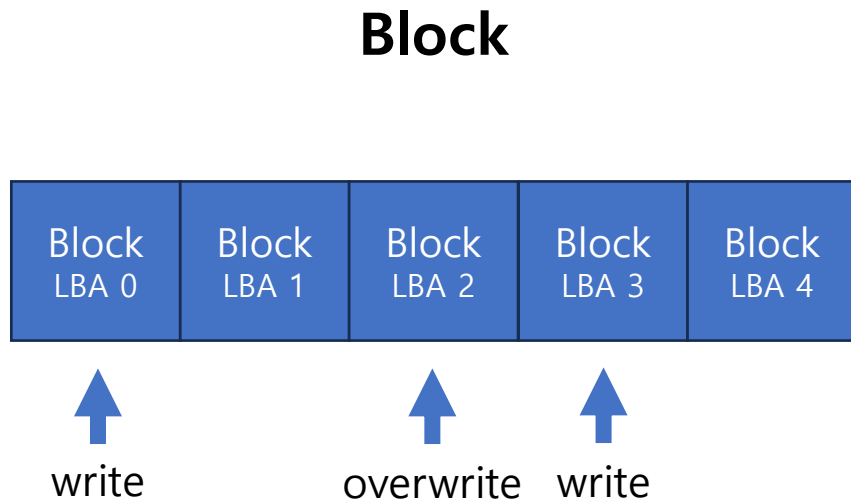
Background



Background



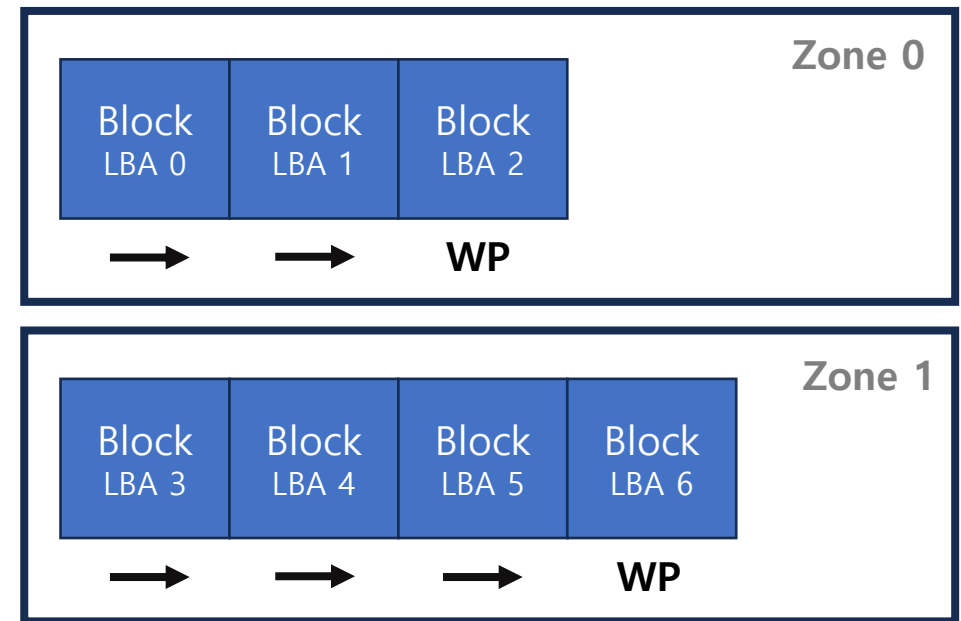
Background



개별 Logical Block을 LBA로 자유롭게 접근

여러 Logical Block을 포함하는 논리 주소 공간의 한 영역

Zone



각 Zone은
Sequential Write
Reset Zone before Reuse

Background

Block SSD

Block Interface + SSD 내부 FTL이 Data Placement, GC 관리

Block Interface

Write in **any order**

Block Block Block Block Block



FTL (Device-side GC + OP)

Disk

ZNS SSD

ZNS Interface + Host가 Zone 단위 Data Placement를 관리

ZNS Interface

Write in **Sequential**

zone
zone



Disk

Experiment Hypothesis

[Block SSD]

- OP가 증가할수록 free space가 증가하여 GC 및 valid-page relocation 부담이 감소
→ Block SSD는 OP 증가에 따라 sustainable write throughput이 증가할 것

[ZNS SSD]

- Host가 zone 단위 placement/reset을 관리하여 device-side GC를 회피
→ ZNS SSD가 Block SSD보다 높은 write throughput을 유지할 것

Original Experiment

SSD Interface	Conv.	Conv.	ZNS
Media Capacity	2 TiB	2 TiB	2 TiB
Host Capacity	1.92 TB	1.60 TB	2 TB
Over-provisioning	7%	28%	0%
Placement Type	None	None	Zones
Max Active Zones	N/A	N/A	14
Zone Size	N/A	N/A	2048 MiB
Zone Capacity	N/A	N/A	1077 MiB

Table 3: Feature summary of the evaluated SSDs

[Hardware / System Environment]

- **Physical Platform:** 모두 동일한 production SSD hardware platform
- Server: Dell R7515
- CPU: 16-core AMD EPYC 7302P
- DRAM: 128GB DDR4-3200
- OS: Ubuntu 20.04
- Kernel: Linux 5.9

Original Experiment

[Workload]

- Write I/O Size: 64KiB
- Write Targets: 4 zones
- Write Pattern: 각 Zone 내부 Sequential Write
- Host Write Target: 0 ~ 1GiB/s
- Block SSD LBA Range: ZNS Zone Capacity와 같은 크기 Range
(동일 Workload 비교 위한 설정 → SSD 내부 차이에 집중)

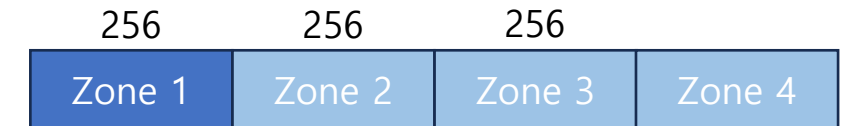
[ZNS / Block SSD 동작]

- Zone 재사용 동작: Zone 재사용 전 **Zone Reset**
- Block 재사용 동작: LBA Range 재사용 전 **TRIM**

[Preconditioning]

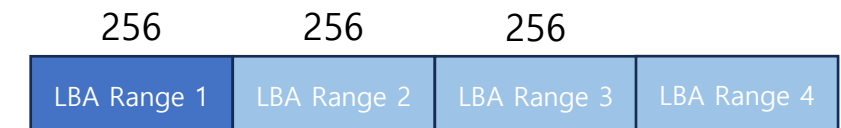
- Steady-state 도달한 상태
- 이후 실험 진행

ZNS SSD



Zone Reset

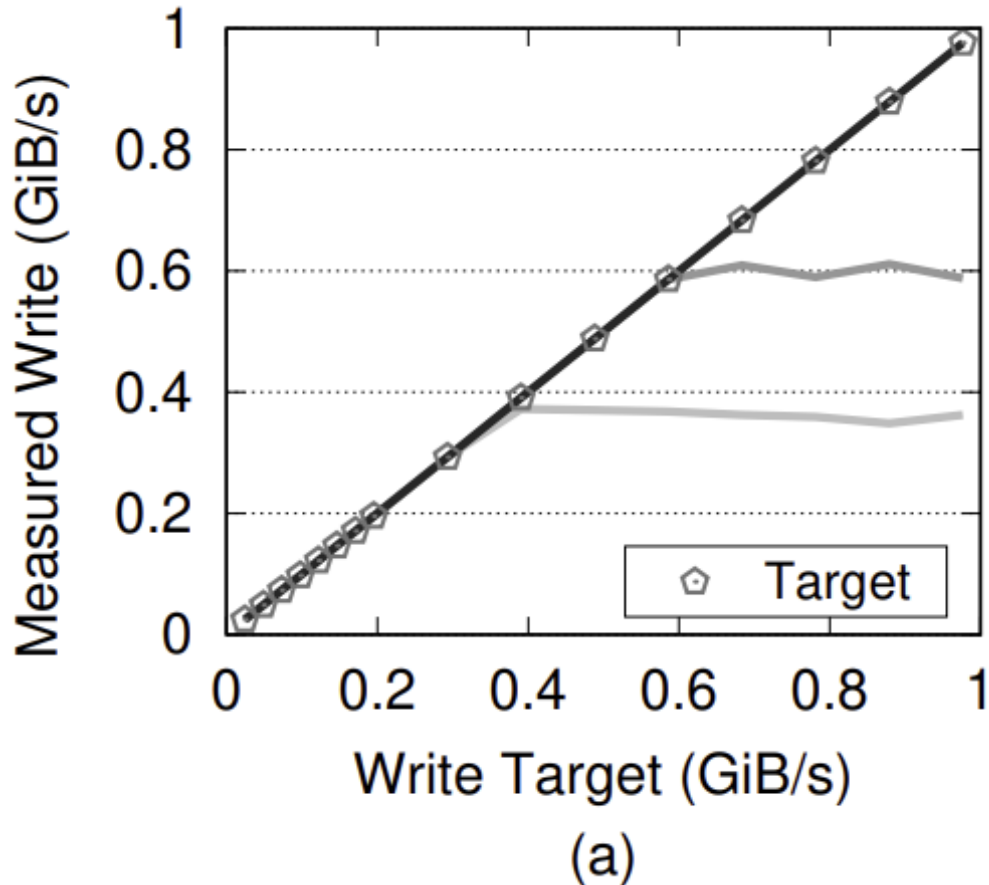
Block SSD



Range TRIM

Original Experiment

목표로 한 Write량에 따른 SSD별 실제 Write 측정량



— ZNS SSD (0% OP)
— Block SSD (7% OP)
— Block SSD (28% OP)

- **Write Target:** Write 목표량
- **Measured Write:** 실제 측정된 Write량
- **Target:** Write Target 따른 실제 Measured Write

- **Block SSD (7% OP):** 약 **0.36~0.38 GiB/s**에서 처리량이 포화
- **Block SSD (28% OP):** 약 **0.6 GiB/s**에서 처리량이 포화
- **ZNS SSD (0% OP):** **1 GiB/s**까지 Target Write를 거의 그대로 따라가며 선형적으로 증가

Block SSD는 OP를 늘리면 GC 부담이 줄어 처리량이 개선
ZNS SSD는 Device-side GC X → 1GiB/s까지 그대로 추종

Reproducing Process

Scaling

기존 실험 환경을
적절히 scaling

Benchmark

실험 진행을 위한
환경 수정

Experiment

FEMU 환경에서
실험 진행

Results

분석 결과 추출 및
그래프 생성

Scaled & Preserved Configuration

	Conv. 7%	Conv. 28%	ZNS	적용 방식
Media Capacity	16 GiB	16 GiB	16 GiB	Scaled
Host Capacity	15 GiB	12.5 GiB	16 GiB	
Zone Size	N/A	N/A	256 MiB	
Zone Capacity	N/A	N/A	256 MiB	
Zone Count	N/A	N/A	64	Derived
Over-provisioning	≈7%	28%	0%	Preserved
Max Active Zones	N/A	N/A	14	
Placement Type	None	None	Zones	

Common Hardware Configuration

[Hardware]

(Size)

- Channel: 8
- LUN/Channel: 8
- Plane/LUN: 1
- Block/Plane: 64
- Page/Block: 256
- Total SSD = 16GiB
- Channel = 2GiB
- LUN/Plane = 256MiB
- Block = 4MiB
- Page = 16KiB

[Latency]

- Read Latency: 32 μ s
- Program Latency: 937.5 μ s
- Erase Latency: 3ms

[GC Threshold Settings]

- ZNS: N/A
- Block
 - Background GC = 90
 - Foreground GC = 97

Setting Reasons

	Conv. 7%	Conv. 28%	ZNS
Media Capacity	16 GiB	16 GiB	16 GiB
Host Capacity	15 GiB	12.5 GiB	16 GiB
Zone Size	N/A	N/A	256 MiB
Zone Capacity	N/A	N/A	256 MiB
Zone Count	N/A	N/A	64
Over-provisioning	≈7%	28%	0%
Max Active Zones	N/A	N/A	14
Placement Type	None	None	Zones

- **Media Capacity: 2 TiB → 16 GiB**
 $8 \times 8 \times 1 \times 64 \times 256 \times 16 \text{ KiB} = 16 \text{ GiB raw}$
- **Host Capacity: 각 OP 비율 고려하여 조정**
- **Zone Size: 2048 MiB → 256 MiB**
 $8Ch \times 8LUN \times 1Plane \times 256Page \times 16KiB = 256MiB$
- **Zone Capacity: 1077 MiB → 256 MiB**
 ZCAP 감소 시 usable capacity 감소
 전체 용량 활용 위해 Zone Size = Zone Capacity

Setting Reasons

	Conv. 7%	Conv. 28%	ZNS
Media Capacity	16 GiB	16 GiB	16 GiB
Host Capacity	15 GiB	12.5 GiB	16 GiB
Zone Size	N/A	N/A	256 MiB
Zone Capacity	N/A	N/A	256 MiB
Zone Count	N/A	N/A	64
Over-provisioning	≈7%	28%	0%
Max Active Zones	N/A	N/A	14
Placement Type	None	None	Zones

- **Zone Count: 64**
 $16 \text{ GiB} / 256 \text{ MiB} = 64 \text{ Zones}$
- **Over-provisioning: 기존 비율 유지**
- **Max Active Zones: 원본 값 14 유지**
 용량 스케일링과 무관한 동시 활성 Zone 제약
 → 원본 값을 유지

Setting Reasons

[ZNSSD Hardware]

- Channel: 8
- LUN/Channel: 8
- Plane/LUN: 1
- Block/Plane: 64
- Page/Block: 256
- Page Size: 16KiB



[ZNSSD 구조를 기준 + BBSSD를 수정]

- 기존 BBSSD Page Size: 4KiB → 16KiB
- 기존 BBSSD Block/Plane: 256 → 64

(ZNS/Block의 Media Capacity를 가능한 한 동일하게 설정)

[Unified Latency]

- Read Latency: 32μs
- Program Latency: 937.5μs
- Erase Latency: 3ms

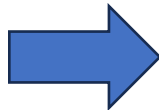
Setting Reasons

[ZNSSD Hardware]

- Channel: 8
- LUN/Channel: 8
- Plane/LUN: 1
- Block/Plane: 64
- Page/Block: 256
- Page Size: 16KiB

[Unified Latency]

- Read Latency: 32 μ s
- Program Latency: 937.5 μ s
- Erase Latency: 3ms



[ZNSSD Latency Model 기준 + BBSSD Latency 수정]

- ZNSSD의 TLC에 맞게 BBSSD 수정
 - Read Latency = 40,000ns → 32,000ns
 - Program Latency = 200,000ns → 937,500ns
 - Erase Latency = 2,000,000ns → 3,000,000ns

[Latency 설정 차이점]

- BBSSD는 사용자 지정 가능
- ZNSSD는 4가지의 Flash Mode 선택

ZNSSD Flash Mode	Features
SLC (1)	너무 빠름 → 일반 SSD 특성 반영 부족
MLC (2)	FEMU ZNSSD에서 사용 불가
TLC (3)	현실적인 성능·밀도 균형
QLC (4)	Program Latency가 매우 큼

Setting Reasons

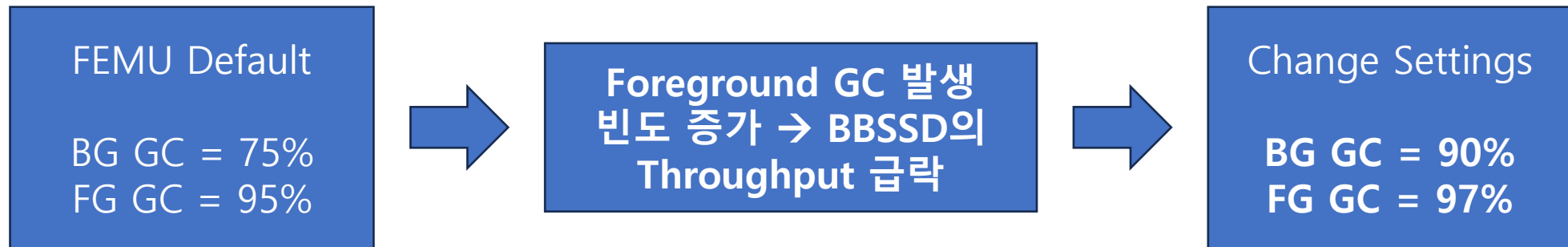
[GC Threshold Settings]

- ZNS: N/A
- Block
 - Background GC = 90
 - Foreground GC = 97

[GC 2가지 동작]

- Background GC → Free Line이 부족해지면 Host I/O 처리 후 GC 수행
- Foreground GC → Free Line이 매우 부족하면 Host Write를 중단하고 GC 수행

- Line: 여러 LUN/Channel의 같은 위치 Block들을 묶어 관리하는 GC 단위



Setting Reasons

[GC Threshold Settings]

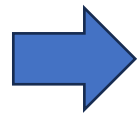
- ZNS: N/A
- Block
 - Background GC = 90
 - Foreground GC = 97

[GC 2가지 동작]

- Background GC → Free Line이 부족해지면 Host I/O 처리 후 GC 수행
- Foreground GC → Free Line이 매우 부족하면 Host Write를 중단함

FEMU Default

BG GC = 75%
FG GC = 95%



Foreground GC
빈도 증가 → BBSS
Throughput 급증

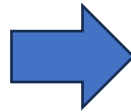
```
femu@fvm:~/zbdbench$ cat result
write_target_mbs,write_avg_mbs
25,25.007254852539063
50,49.20389268652344
75,75.00711262988281
100,100.0262939453125
125,125.04904030273437
150,148.8758568720703
175,108.205078125
200,85.0129972578125
300,79.69724386328124
400,74.269903921875
500,72.1905384970703
600,70.5858368515625
700,69.856072640625
800,68.90001898828125
900,69.42941351953125
1000,68.30595160546875
```

위치
한위

Environment Settings

run-zns-fig5a.sh

```
SSD_SIZE_MB=4096  
NUM_CHANNELS=8  
NUM_CHIPS_PER_CHANNEL=4  
NUM_PLANES_PER_CHIP=2  
NUM_BLOCKS_PER_CHIP=32  
# SLC:1 MLC:2 TLC:3 QLC:4  
# MLC is not allowed  
FLASH_TYPE=4
```



```
SSD_SIZE_MB=16384  
NUM_CHANNELS=8  
NUM_CHIPS_PER_CHANNEL=8  
NUM_PLANES_PER_CHIP=1  
NUM_BLOCKS_PER_CHIP=64  
# SLC:1 MLC:2 TLC:3 QLC:4  
# MLC is not allowed  
FLASH_TYPE=3
```

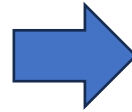
Environment Settings

run-bbssd-fig5a-**op7**-gc90-97.sh

```
# FEMU Default
secs_per_pg=8
blks_per_pl=256
ssd_size=12288

pg_rd_lat=40000
pg_wr_lat=200000
blk_er_lat=2000000

gc_thres_pcent=75
gc_thres_pcent_high=95
```



```
# Experiment
secs_per_pg=32
blks_per_pl=64
ssd_size=15360

pg_rd_lat=32000
pg_wr_lat=937500
blk_er_lat=3000000

gc_thres_pcent=90
gc_thres_pcent_high=97
```

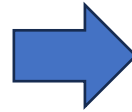
Environment Settings

run-bbssd-fig5a-**op28**-gc90-97.sh

```
# FEMU Default
secs_per_pg=8
blks_per_pl=256
ssd_size=12288

pg_rd_lat=40000
pg_wr_lat=200000
blk_er_lat=2000000

gc_thres_pcent=75
gc_thres_pcent_high=95
```



```
# Experiment
secs_per_pg=32
blks_per_pl=64
ssd_size=12800

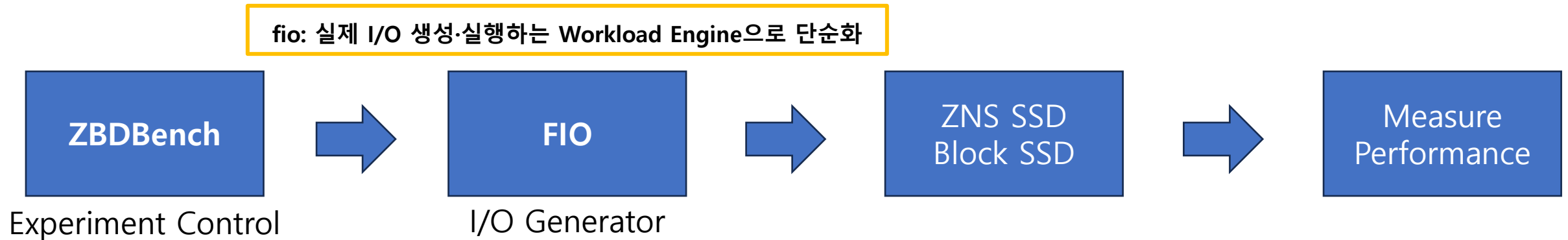
pg_rd_lat=32000
pg_wr_lat=937500
blk_er_lat=3000000

gc_thres_pcent=90
gc_thres_pcent_high=97
```

ZBDBench

ZBDBench: Western Digital에서 공개한 Zoned Block Device Benchmark Suite

- ZNS / Conventional Block Device 성능 평가
 - FIO 기반 I/O workload 생성 및 실행
- 실제로 동일 논문의 5.2 RocksDB 실험을 재현하기 위한 Benchmark을 포함하고 있음 (zns_atc_paper_eval)



기존 ZBDBench 구조를 활용하여 현재 실험에 적합한 Benchmark로 개조
→ Figure 5(a) 전용 workload 실행 및 결과 수집에 사용

ZBDBench

fio_fig5a.py

- ZBDBench의 원본 fio_zone_mixed.py를 실험 목적 및 환경에 맞게 Guest OS에서 수정

[수정사항]

- 실험 조건에 맞는 워크로드를 설정
 - Write-only Workload
 - Synthetic Zone Size, Max Open Zone Value..
 - Preconditioning
 - Queue Depth

ZBDBench

1. Read workload 제거 → Write-only Workload 생성

```
for s in rates:
    mixs_param += (
        f" --name=mix_{s}_w"
        f" --rate={s}m"
        f" --bs=64k"
        ...
        f" --name=mix_{s}_r"      # ← 삭제
        f" --rw=randread"        # ← 삭제
        f" --bs=4k"              # ← 삭제
        f" --ramp_time=30"       # ← 삭제
        ...
    )
```

25 MiB/s Write + Random Read
50 MiB/s Write + Random Read
75 MiB/s Write + Random Read
...



25 MiB/s Write
50 MiB/s Write
75 MiB/s Write
...

ZBDBench

2. 워크로드 설정: Write Size, Target Zone, Synthetic Zone

- **Rate Sweep 설정** (Target Write Rate를 단계적으로 증가시키며 실제 Throughput 측정)

```
max_open_zones = 4

extra = '--zonesize=262144k' # Block SSD

init_param = (
    f" --direct=1"
    f" --zonemode=zbd"
    f" --rw=randwrite"
    f" --bs=64k"
    f" --iodepth=64"
)
```

Parameter	Setting
Write Size	64KiB
Max Open Zone	4
Synthetic Zone	256MiB

```
for s in [
    25, 50, 75, 100, 125, 150, 175, 200,
    300, 400, 500, 600, 700, 800, 900, 1000
]:
    mixs_param += (
        f" --name=mix_{s}_w"
        f" --wait_for_previous"
        f" --rate={s}m"
        f" --runtime=180"
        f" --time_based"
    )
```

ZBDBench

3. Preconditioning

- 전체 device reset → 2x Host-visible Capacity Write → Preconditioning

```
def setup(...):  
    ...  
    self.discard_dev(dev)
```

```
if is_dev_zoned(dev):  
    zonecap = 100
```

```
io_size = int(  
    ((self.get_dev_size(dev) * zonecap) / 100) * 2  
)  
  
prep_param = (  
    f"--name=prep"  
    f" --io_size={io_size}k"  
)
```

Device	Host Cap.	Prep Write
ZNS	16GiB	32GiB
Block 7%	15GiB	30GiB
Block 28%	12.5GiB	25GiB

ZBDBench

4. Queue Depth(QD) 설정 & 결과 CSV

- Host submission queue가 먼저 bottleneck 문제 발생
- **QD = 64 적용 (Both ZNS, Block)**
- 결과 출력 단순화 (JSON에서 Target Write BW / Measured Write BW만 CSV로 추출)

```
max_open_zones = 4

extra = '--zonesize=262144k' # Block SSD

init_param = (
    f" --direct=1"
    f" --zonemode=zbd"
    f" --rw=randwrite"
    f" --bs=64k"
    f" --iodepth=64"
)
```

```
w.writerow([
    'write_target_mbs',
    'write_avg_mbs'
])

w.writerows(csv_data)
```

FIO

FIO의 Range Reset 문제 (Block SSD)

[기존 방식]

- LBA를 Zone Capacity 크기의 Range로 분할 → **Range 재사용 전 TRIM**

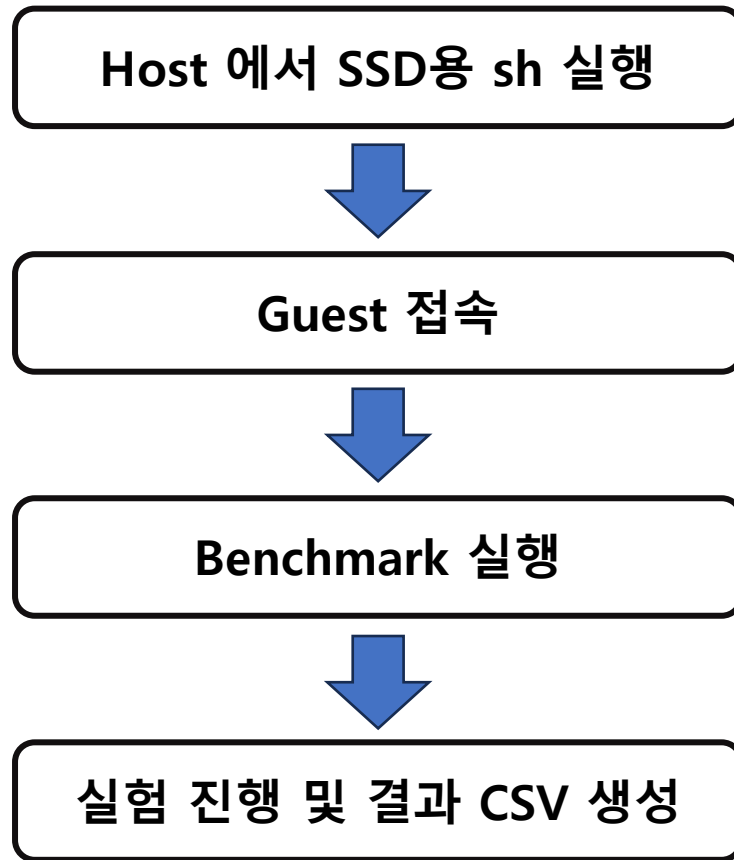
[FIO 처리 방식]

- FIO의 Synthetic 방식 → Range 재사용 시 내부 **Write Pointer만 Reset** → **실제 TRIM 없음**

기본 fio에서는 Range 재사용 시, 실제 TRIM이 발생하지 않아 논문의 Block SSD 조건과 불일치

Solution: Synthetic Reset 시 실제 TRIM/Discard 발생하도록 os_trim()을 호출하도록 수정

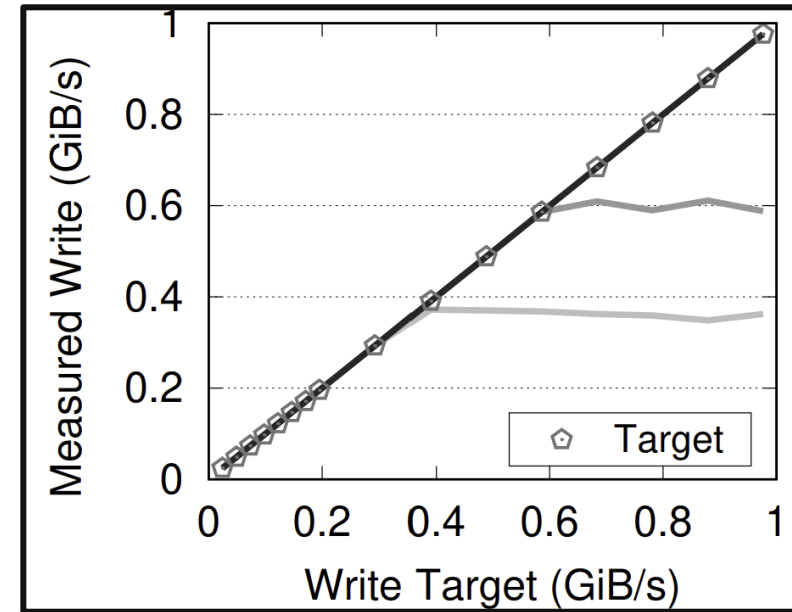
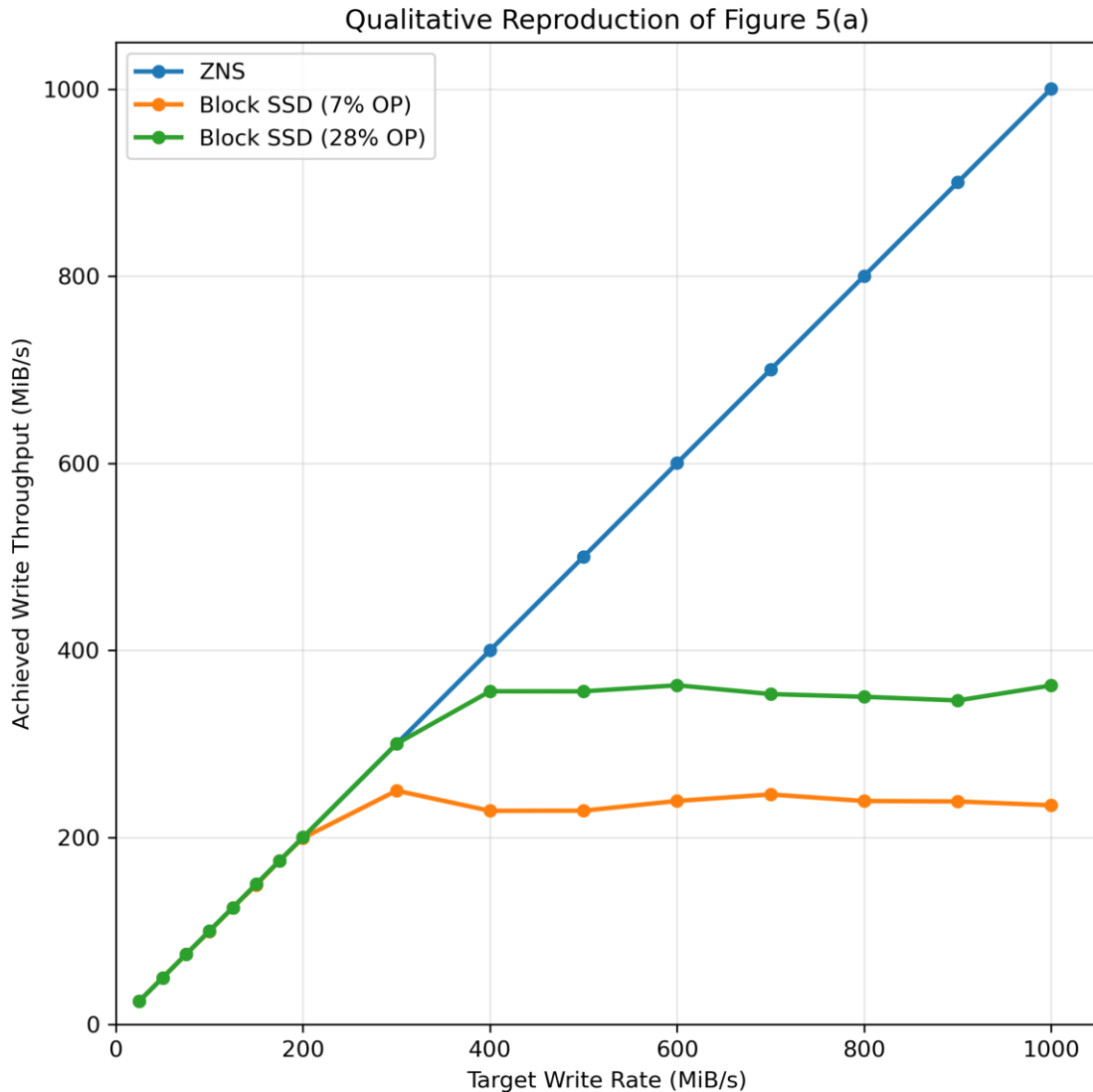
Experiment Process



ZNS SSD, Block 7%, Block 28% 하나씩 .sh 파일 실행
(총 3번 과정 반복)

각 실험 결과 CSV 바탕으로 Graph 생성

Results



[실험 결과]

- Block 7% $\approx$ 240 MiB/s
- Block 28% $\approx$ 350~360 MiB/s
- ZNS $\approx$ 1000 MiB/s

절대적 성능값은 다르지만 성능 순서와 처리량 포화 경향은 재현

Limitation

Experiment Limitations

- **FTL / GC 차이**
실제 Block SSD의 FTL/GC는 비공개 → FEMU BBSSD의 **Page-level Mapping + Global WP + Line-based GC**로 대체
- **Zone Geometry 차이**
Paper: **2048 / 1077 MiB** → FEMU: **256 / 256 MiB**로 단순화
- **미공개 Parameter 설정**
QD, NAND Timing, Preconditioning, Runtime 등은 실험 환경에 맞게 설정 (*QD64, TLC, 2× host-visible Capacity, 180 s/point*)
- **GC Threshold 조정**
Small FEMU SSD의 GC Thrashing 완화를 위해 **75/95 → 90/97**

Interpretation

- 절대적인 Throughput 재현보다 Block 7% OP < Block 28% OP < ZNS의 성능 재현을 목표로 함

Conclusion

Hypothesis

- **Block SSD:** OP 증가 → GC/Relocation pressure 감소 → **Sustained Write Throughput ↑**
- **ZNS SSD:** Device-side GC 회피 → **Block SSD 대비, high sustainable Throughput**

Result

- **Block 7% OP** $\approx$ 240 MiB/s
- **Block 28% OP** $\approx$ 350~360 MiB/s
- **ZNS** $\approx$ 1000 MiB/s

Block 7% < Block 28% < ZNS

Conclusion

- OP 증가에 따른 Block SSD의 성능 향상 확인
- ZNS는 Device-side GC 없음 → **주어진 처리량 유지**
- **Figure 5(a)의 성능 순서와 경향 재현에 성공**

Thank you

Appendix

FIO의 Range Reset 문제 (Block SSD)

default:

```
ret = os_trim(f, offset, length);  
if (ret) {  
    td_verror(td, ret, "discarding synthetic zone failed");  
    return -ret;  
}  
break;
```

Synthetic Range Reset



os_trim(offset, length)



BLKDISCARD



FEMU BBSSD TRIM

Validation: strace를 통해 Range Reset 횟수와
BLKDISCARD 발생 횟수가 일치함을 확인